# Lecture 5: Power BJT

EEE 339 Power Electronics · Complete study notes with previous-year solutions · Source: *class slides/Lecture 05 Power BJT.pdf* (Rashid ch. 4)

**Exam priority (from the 2019 to 2024 papers):**

1. **Switching characteristics** (t_d, t_r, t_s, t_f with waveforms). Asked in 2022 Q4b [6].
2. **Second breakdown**, and why it occurs in BJTs and thyristors but not MOSFETs. Asked in 2024 Q3b [7].
3. **Parallel operation**: why BJTs and SCRs cannot be paralleled easily but MOSFETs can. Asked in 2024 Q1d [5].
4. **Comparison tables** (BJT vs MOSFET vs IGBT, and others). Asked in 2021 Q1a [7], 2022 Q4a [5], 2023 Q6d, 2019 TT-1.
5. Constructional features (2023 Q6c), modes / quasi-saturation, SOA, base drive.

Expected weight in your 60-mark paper: one theory sub-part of 3 to 6 marks in Part A (devices).

## 1. Introduction and basics

- The BJT was the traditional power switch. IGBTs and MOSFETs have now largely replaced it, but BJTs are still used where **cost** is the main concern.
- It is a **current-controlled**, **bipolar** (minority-carrier) device: a base current I_B is needed to keep it on (I_C = βI_B in the active region).

### Basics of a power BJT (slide 2)

- Larger current and power handling than a small-signal BJT.
- Higher voltage blocking in the OFF state; higher current in the ON state.
- **Vertically oriented** structure (current flows from top to bottom).
- Emitter is very heavily doped compared with the base; even so, the current gain **β is low** (typically 5 to 20).
- Large metal body (TO-3 style case) to dissipate heat.

## 2. Structure (constructional features)

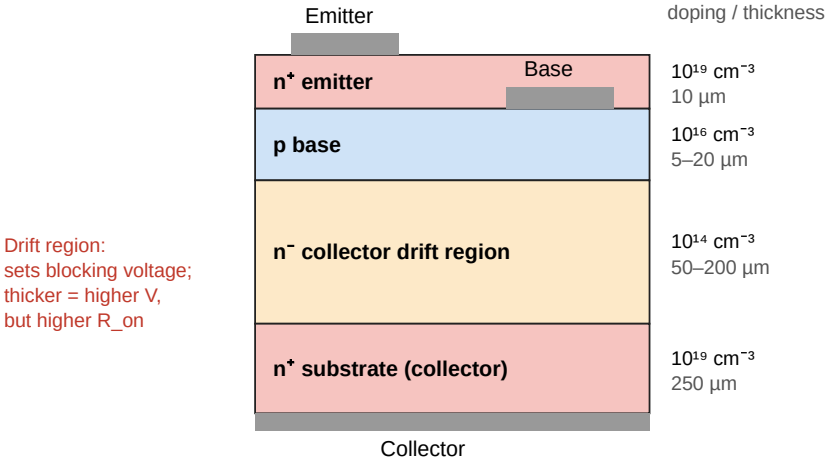



Fig. 1: Vertical n⁺-p-n⁻-n⁺ structure of a power BJT (doping and thickness from the slide figure).

- **Four layers:** n⁺ emitter, p base, **n⁻ collector drift region**, n⁺ substrate (collector contact).
- **Drift region** (lightly doped, thick): it increases the blocking (voltage) capacity. Thicker drift gives a higher breakdown voltage but a higher on-state resistance and causes the quasi-saturation region.

- **Vertical structure:** maximises the cross-section area for current, so on-resistance and thermal resistance are low.
- **Interdigitated emitter and base** (many narrow emitter fingers) to reduce emitter current crowding.
- Low β, so a **Darlington** connection is often used to raise the gain.

## 3. Operating modes and output characteristics

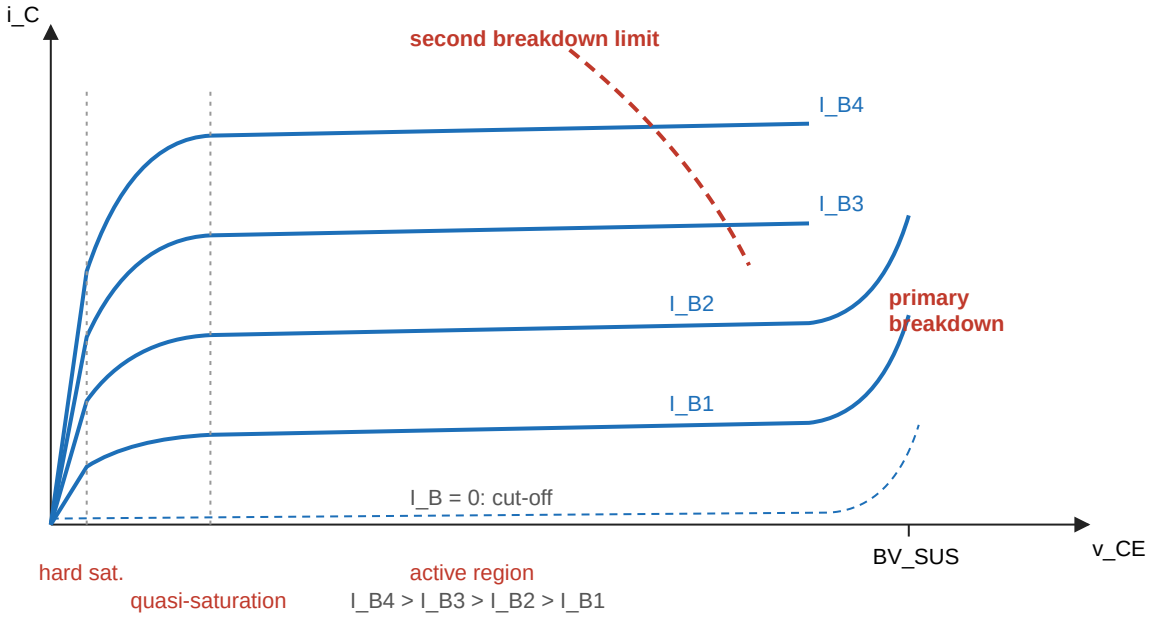



Fig. 2: Output characteristics of a power BJT showing hard saturation, quasi-saturation, active region, cut-off, primary breakdown and the second-breakdown limit.

| Mode | Junction bias | Notes |
|---|---|---|
| **Cut-off** | B-E reverse, C-B reverse | I_C ≈ 0 (leakage); switch OFF. |
| **Active** | B-E forward, C-B reverse | I_C = βI_B; v_CE large, so loss is high. In switching, cross it **as fast as possible**. |
| **Quasi-saturation** | B-E forward, C-B (just) forward | Extra region of **vertical** power BJTs: the drift region is not yet flooded with carriers, so v_CE is moderate. Less stored charge, so faster turn-off. |
| **Hard saturation** | B-E forward, C-B forward | Drift region fully conductivity-modulated; v_CE = V_CE(sat) is lowest (< 0.7 V region per slide). Lowest conduction loss, but most stored charge. |

In switching, the BJT is used only in **saturation (ON)** and **cut-off (OFF)**; in both, the power loss v_CE·i_C is low.

## 4. Breakdown: primary and second breakdown

### Primary breakdown

- Normal **avalanche breakdown of the collector-base junction**, with a large current.
- **BV_SUS** (sustaining voltage) is the limit: the maximum v_CE the transistor can sustain while carrying collector current.
- **BV_SUS < BV_CEO < BV_CBO**. BV_CEO and BV_CBO are the standoff voltages with zero or negative base current.

### Second breakdown (secondary breakdown)

- A failure mode seen as a **sudden (precipitous) drop of v_CE at large currents**.
- **Cause:** power dissipation is not uniform across the chip but concentrated in small regions. The local temperature rises quickly, and a BJT's resistance **falls** as temperature rises (negative temperature coefficient). So the hot spot conducts even more current, gets hotter still: **thermal runaway**.
- The **hot spots** can melt and recrystallise the silicon, destroying the device.

```
hot spot → lower resistance there → more current → more heat → hotter spot → … → melting
```

### How to avoid second breakdown (slide)

1. Keep power dissipation under control.
2. Use a controlled rate of change of base current during turn-off.
3. Use protective snubber circuits.
4. Keep the switching trajectory within the SOA boundaries.

## 5. Safe operating area (SOA)

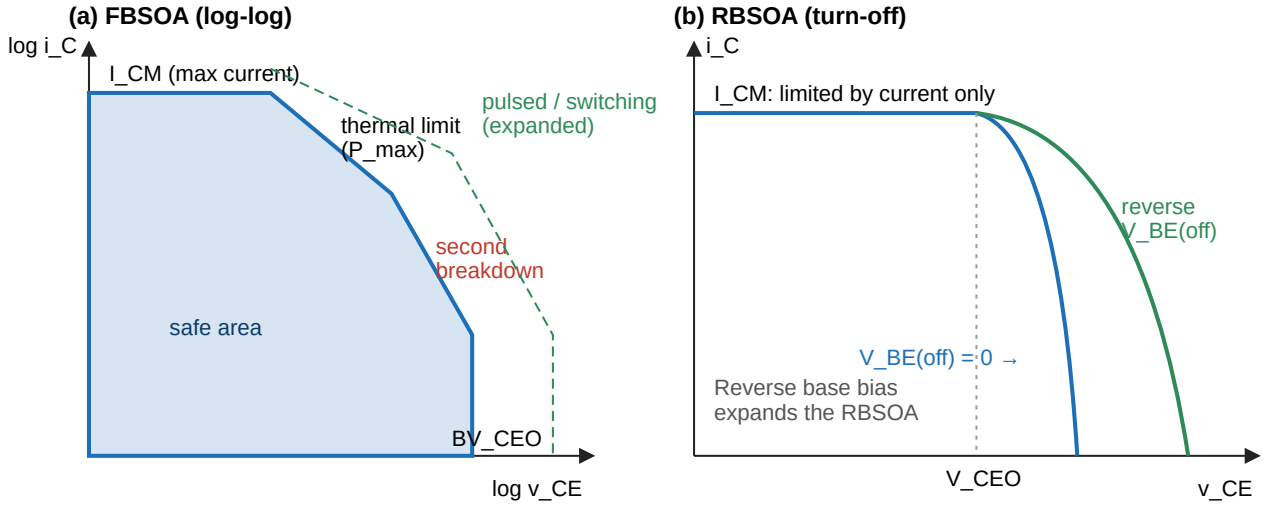



Fig. 3: (a) Forward-bias SOA with its four boundaries. (b) Reverse-bias SOA at turn-off.

### FBSOA (forward-bias SOA, also simply called SOA)

- Boundaries: **I_CM** (maximum current), the **thermal limit** (maximum power dissipation), the **second-breakdown limit**, and the maximum voltage **BV_CEO**.
- The area **expands** for pulsed (switching) operation.

### RBSOA (reverse-bias SOA, during turn-off)

- Inductive loads give higher peak energy at turn-off than resistive loads, so second breakdown can happen if the RBSOA is exceeded.
- A **reverse base current expands the RBSOA**.
- Below V_CEO, the safe area does not depend on the reverse bias V_EB; it is limited only by I_C. Above V_CEO, the collector current must be controlled depending on the reverse-bias voltage.
- Temperature derates the SOA.

### Improving SOA capability

SOA is set by peak power handling, which depends on current density and thermal density. Keep both low: a **wide base** limits current density, and **good heat sinking under the collector** lets

the device withstand high peak power. Spreading power over a large silicon area gives the highest SOA. Failure starts from a hot spot at the weakest point.

## 6. Switching characteristics

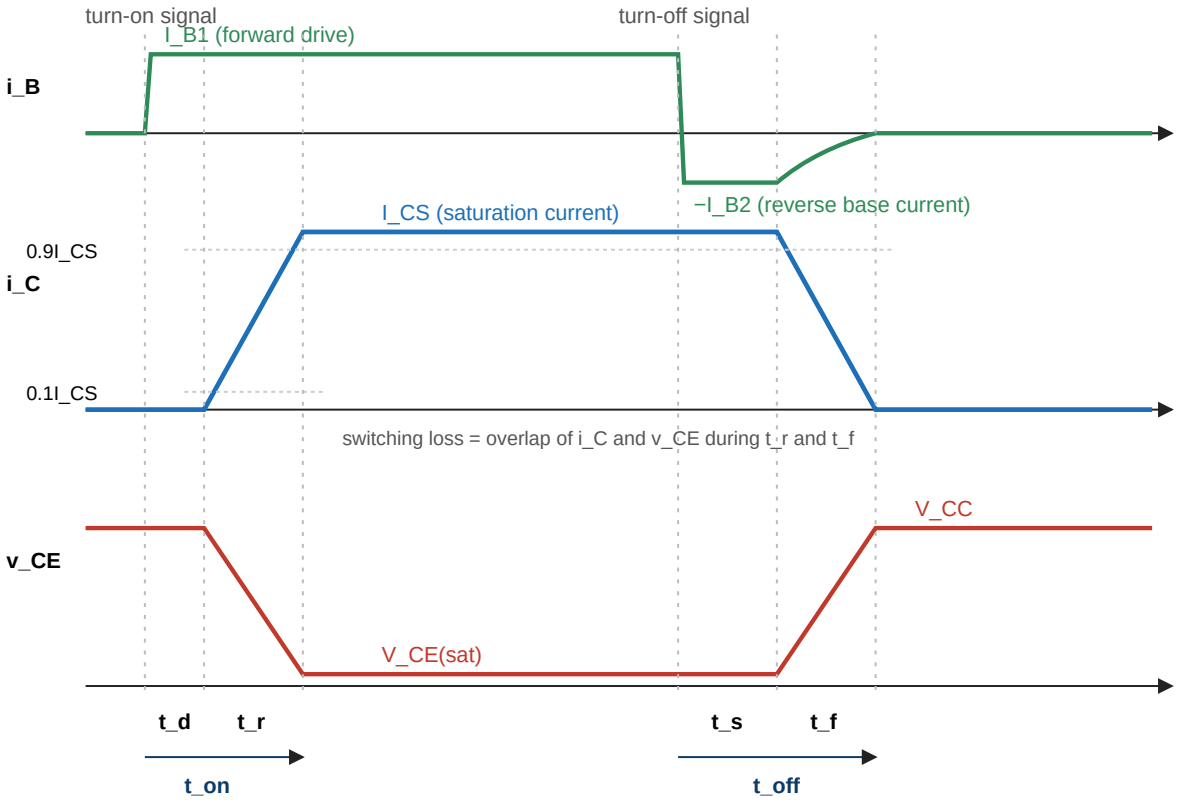



Fig. 4: Switching waveforms of a power BJT (resistive load): base current, collector current and collector-emitter voltage.

| Time | Meaning |
|---|---|
| **Delay time t_d** | From the base drive until i_C reaches 10%. Time to charge the **base-emitter junction capacitance** to forward bias. Reduced by a **larger base current with a high slope**. |
| **Rise time t_r** | i_C rises from 10% to 90% of I_CS (index "r" = 10 to 90%). |
| **Turn-on time** | **t_on = t_d + t_r** |
| **Storage time t_s** | After the turn-off signal, the time needed to **neutralise the minority carriers stored in the base and collector**. i_C stays almost constant. The most important BJT parameter; the main limit on switching speed. |
| **Fall time t_f** | i_C falls from 90% to 10%: crossing from saturation to cut-off. Reduced by a **negative base current** and by keeping the device in **quasi-saturation** (less stored charge). |
| **Turn-off time** | **t_off = t_s + t_f** |

**Switching losses** occur during t_r and t_f, when v_CE and i_C are both large at the same time. Storage time and switching losses are the key issues with power BJTs.

### Base drive optimisation

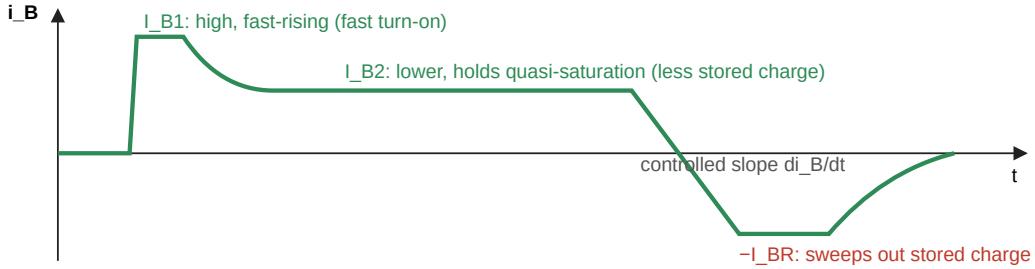



Fig. 5: Ideal base-drive current waveform.

- **I_B1**: high forward base current with a fast rise, so the BJT turns on quickly (small t_d and t_r).
- **I_B2**: lower holding current that keeps the BJT in **quasi-saturation**: conduction loss stays low and excess base charge is avoided.
- **−I_BR** with a controlled slope: a reverse base current sweeps out the stored charge, reducing t_s and t_f (and expanding the RBSOA).

## 7. Applications

SMPS · power amplifiers · DC-to-AC inverters · relays · power control circuits.

## 8. Previous-year questions with full solutions

2022 Q4(b) **Q. Explain the switching characteristics of a BJT. [6]**

**Answer.** Draw Fig. 4 (i_B, i_C and v_CE on the same time axis) and mark t_d, t_r, t_s, t_f.

- **Turn-on:** when the base current I_B1 is applied, i_C does not rise at once. During the **delay time t_d** the base-emitter junction capacitance charges up to forward bias. Then i_C rises from 10% to 90% of I_CS during the **rise time t_r**, while v_CE falls to V_CE(sat). **t_on = t_d + t_r**.
- **Turn-off:** when the base current is reversed (−I_B2), the excess minority carriers stored in the base and collector must first be removed. During this **storage time t_s**, i_C stays almost constant. Then i_C falls from 90% to 10% during the **fall time t_f** while v_CE rises to V_CC. **t_off = t_s + t_f**.
- **Losses:** during t_r and t_f both v_CE and i_C are large, so most switching loss occurs there.
- **Improving switching:** a large, fast-rising I_B1 reduces t_d and t_r; holding the device in quasi-saturation (I_B2) reduces stored charge; a negative base current reduces t_s and t_f.

2024 Q3(b) **Q. What is secondary breakdown? Explain why secondary breakdown occurs in BJT and thyristors but not in MOSFETs. [7]**

**Secondary breakdown** is a destructive failure of a power transistor at high current and voltage, seen as a sudden collapse of v_CE while the current is large. It is caused by **localised thermal runaway** (hot spots), not by the normal avalanche of the junction (that is primary breakdown). Mark the second-breakdown limit on the output characteristics or SOA (Fig. 2 or 3a).

**Why in BJTs and thyristors:**

- They are **bipolar (minority-carrier) devices**. Their on-state resistance has a **negative temperature coefficient**: as temperature rises, more carriers are generated, so resistance falls.
- Current is never perfectly uniform. A slightly hotter region therefore carries **more** current, which heats it further (**current crowding** / filamentation).
- This positive feedback gives **local thermal runaway**; the hot spot melts and recrystallises the silicon and the device is destroyed. In thyristors the regenerative (latching) bipolar action makes the current filament worse.

**Why not in MOSFETs:**

- A MOSFET is a **majority-carrier (unipolar)** device with no minority-carrier storage.
- Its R_DS(on) has a **positive temperature coefficient**: carrier mobility falls as temperature rises, so resistance rises.
- A hotter region therefore carries **less** current, and current shifts to cooler regions. The current spreads evenly over the chip (self-stabilising), so no hot spot or thermal runaway can form.

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BJT: T↑ → R↓ → I↑ → T↑↑ (runaway)   |   MOSFET: T↑ → R↑ → I↓ → T stable
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2024 Q1(d) **Q. Explain why BJT and SCR cannot be operated in parallel, whereas MOSFETs can operate in parallel. [5]**

When devices are paralleled, they must **share the current equally**. Whether they do depends on the temperature coefficient of their on-state resistance.

- **BJT:** negative temperature coefficient (V_CE(sat) falls as temperature rises). The transistor with slightly lower V_CE(sat) takes more current, heats up, and its V_CE(sat) falls further, so it takes even more. This **current hogging** leads to thermal runaway of one device. Paralleling needs matched devices and **emitter ballast resistors**.
- **SCR:** also a bipolar device with negative temperature coefficient. In addition, SCRs have different on-state voltages and different turn-on times: the first one to latch takes the whole current. Paralleling needs series resistors or inductors and simultaneous, strong gate pulses.
- **MOSFET:** R_DS(on) has a **positive temperature coefficient**. The device carrying more current heats up, its resistance rises and its current falls. Current sharing is **automatic**, so MOSFETs can be paralleled easily; only matched gate drive (small gate resistors) is needed.

2023 Q6(c) **Q. Mention the constructional features of a power BJT. [part of 8]**

Draw Fig. 1, then list:

1. **Vertical structure** $n^+$-p-$n^-$-$n^+$: the collector is at the bottom, so the current-carrying area is large.
2. Heavily doped $n^+$ emitter (≈$10^{19}$ $cm^{-3}$, ≈10 µm), p base (≈$10^{16}$, 5–20 µm).
3. **Lightly doped $n^-$ collector drift region** (≈$10^{14}$, 50–200 µm): gives the high blocking voltage; causes quasi-saturation.
4. $n^+$ substrate (≈250 µm) forms the collector contact.
5. Interdigitated base-emitter fingers to avoid emitter current crowding.
6. Low current gain β (5 to 20); Darlington pairs used for higher gain.
7. Large metal case / heat-sink mounting to dissipate heat.

2022 Q4(a) **Q. Compare between BJT and MOSFET as a switch. [5]**

| Feature | Power BJT | Power MOSFET |
|---|---|---|
| Control | Current-controlled (needs continuous base current) | Voltage-controlled (almost no gate current) |
| Input impedance | Low | Very high |
| Drive circuit | Complex, high drive power | Simple, very low drive power |
| Carriers | Bipolar (minority carriers): storage time | Unipolar (majority carriers): no storage time |
| Switching speed | Slow (up to about 10 kHz) | Very fast (hundreds of kHz to MHz) |
| On-state loss | Low V_CE(sat) | R_DS(on); high for high-voltage devices |
| Temperature coefficient | Negative: thermal runaway | Positive: stable |
| Second breakdown | Yes | No |
| Paralleling | Difficult (current hogging) | Easy (automatic sharing) |
| Cost / ratings | Cheap; medium voltage and current | Higher cost; best at low voltage |

2021 Q1(a) **Q. Differentiate among power BJT, MOSFET and IGBT in terms of power handling capacity, input impedance, switching speed, on-state resistance and blocking voltage. [7]**

| Parameter | Power BJT | Power MOSFET | IGBT |
|---|---|---|---|
| Power handling capacity | Medium | Low to medium | **High** |
| Input impedance | **Low** (current-driven) | **Very high** (insulated gate) | **Very high** (MOS gate) |
| Switching speed | Slow (≈ up to 10 kHz) | **Fastest** (≈ 100 kHz to MHz) | Medium (≈ 20 to 50 kHz) |
| On-state resistance / drop | Low (V_CE(sat) ≈ 1–2 V) | **High** at high voltage (R_DS(on) grows with voltage rating) | **Low** (conductivity modulation) |
| Blocking voltage | Medium (≈ 1–1.5 kV) | Low to medium (≈ up to 1 kV) | **High** (≈ up to 6.5 kV) |

The ratings are approximate typical values; the examiner marks the comparison (low/medium/high), not exact numbers. Summary: the IGBT combines the MOSFET's high input impedance with the BJT's low on-state drop.

2019 TT-1 Q2 **Q. Tabulate the similarity and dissimilarity between power BJT and IGBT in terms of power handling capacity, switching speed, on-state resistance and blocking voltage. [4]**

**Similarities:** both are three-terminal, controllable (turn-on and turn-off), unidirectional switches; both conduct by bipolar action (minority carriers), so both have a low on-state voltage drop; both have a vertical structure with an $n^-$ drift region.

| Parameter | Power BJT | IGBT |
|---|---|---|
| Power handling | Medium | High |
| Switching speed | Slow (storage time) | Faster than BJT (tail current limits it) |
| On-state resistance | Low | Low (similar to BJT) |
| Blocking voltage | Medium (≈ 1–1.5 kV) | High (≈ up to 6.5 kV) |
| Drive | Current-driven, high drive power | Voltage-driven, very low drive power |

2023 Q6(d) **Q. Compare the following devices in terms of power capacity and switching speed: BJT, MOSFET, GTO, IGBT, MCT. [4]**

| Device | Power capacity | Switching speed |
|---|---|---|
| GTO | **Highest** (≈ 6 kV, 6 kA) | **Slowest** (≈ 1 kHz) |
| IGBT | High | Medium-fast (≈ 20–50 kHz) |
| MCT | Medium-high | Medium-fast (≈ 20 kHz) |
| BJT | Medium | Slow-medium (≈ 10 kHz) |
| MOSFET | **Lowest** | **Fastest** (≈ 100 kHz – 1 MHz) |

Order of power: **GTO > IGBT > MCT > BJT > MOSFET**. Order of speed: **MOSFET > IGBT ≈ MCT > BJT > GTO**.

## 9. Practice questions (likely next) with short answers

Predicted **Q. What is quasi-saturation? Why does it occur only in power BJTs?**

A region between the active region and hard saturation where both junctions are (just) forward biased but the lightly doped $n^-$ drift region is not yet fully flooded with carriers, so v_CE is moderate. It occurs because power BJTs have the vertical $n^-$ drift region; small-signal BJTs do not. Operating there reduces stored charge and speeds up turn-off.

Predicted **Q. Draw and explain the FBSOA and RBSOA of a power BJT.**

Draw Fig. 3. FBSOA: limited by I_CM, the thermal (P_max) line, the second-breakdown line and BV_CEO; it expands for pulsed operation. RBSOA: the turn-off limit; below V_CEO only I_C limits it; above V_CEO it depends on the reverse base bias, which expands it. Inductive turn-off is the critical case.

Predicted **Q. Explain the modes of operation of a power BJT. / Explain primary breakdown and BV_SUS.**

Use the table in Section 3 and Fig. 2. Primary breakdown = avalanche of the C-B junction; BV_SUS is the maximum v_CE with collector current flowing; BV_SUS < BV_CEO < BV_CBO.

Predicted **Q. How can the base drive improve the switching performance of a BJT? / How can second breakdown be avoided?**

Base drive: I_B1 (high, fast) for fast turn-on; I_B2 to hold quasi-saturation; −I_BR with controlled slope to remove stored charge (Fig. 5). Second breakdown: control power dissipation, control di_B/dt at turn-off, use snubbers, keep the trajectory inside the SOA.

## 10. One-look revision sheet

| Item | Remember |
|---|---|
| Structure | $n^+$ emitter / p base / **$n^-$ drift** / $n^+$ substrate; vertical; drift sets blocking voltage |
| Modes | Cut-off · Active · **Quasi-saturation** · Hard saturation |
| Breakdown | Primary = C-B avalanche (BV_SUS < BV_CEO < BV_CBO) · Second = hot spot thermal runaway |
| Avoid second breakdown | Limit power · control di_B/dt at turn-off · snubbers · stay in SOA |
| SOA | FBSOA: I_CM, thermal, second breakdown, BV_CEO · RBSOA: turn-off, expanded by reverse base bias |
| Switching | **t_on = t_d + t_r · t_off = t_s + t_f**; t_s (stored charge) is the main limit |
| Base drive | I_B1 high → I_B2 hold (quasi-sat) → −I_BR sweep out |
| Temperature coefficient | BJT/SCR negative: runaway, no easy paralleling · MOSFET positive: stable, easy paralleling |
| Applications | SMPS, power amplifiers, inverters, relays, power control |